PROJECT DOCUMENTATION

Three-Tier Architecture Implementation in Microsoft Azure

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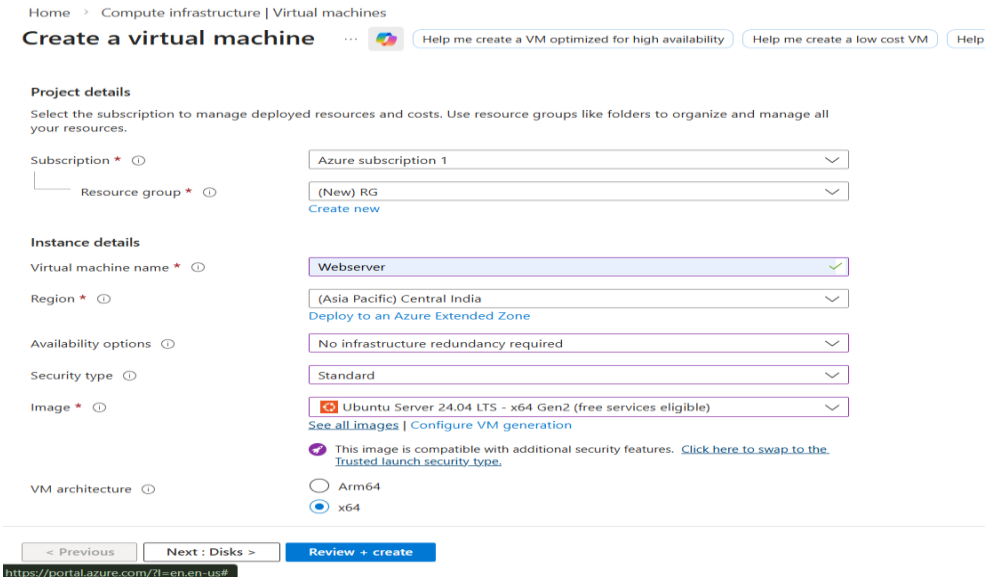
BATCH : 31

Introduction to 3-Tier Architecture

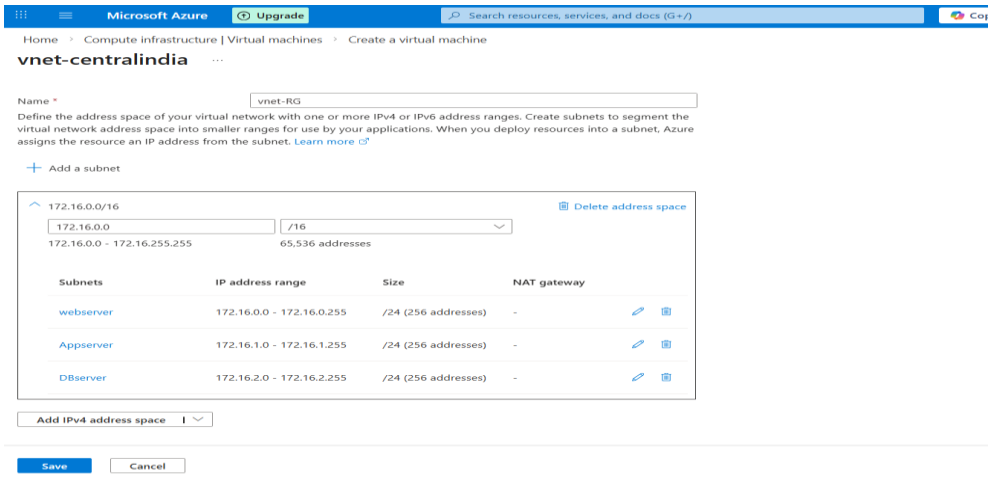
Three-Tier Architecture is a client-server architecture pattern that separates the application into three layers: Web Layer, Application Layer, and Database Layer. This architecture improves security, scalability, and performance.

Implementation Steps

Step 1: Created a Resource Group (RG) in Central India region.
Selected Availability as No Infrastructure Redundancy and Security type as Standard.

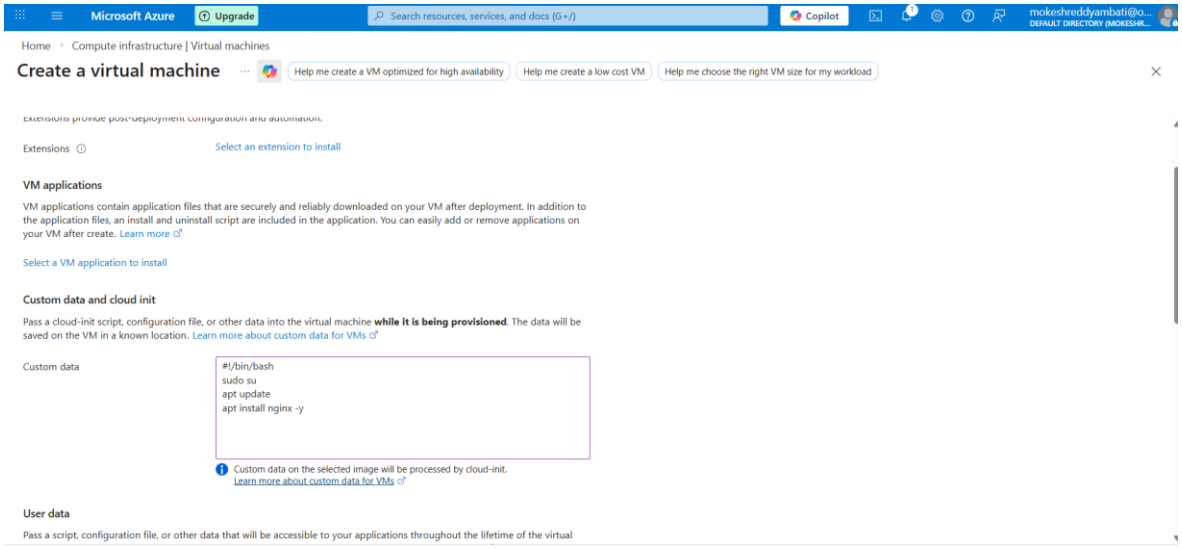


Step 2: Created Virtual Network and three subnets:
- Web Subnet
- App Subnet
- DB Subnet

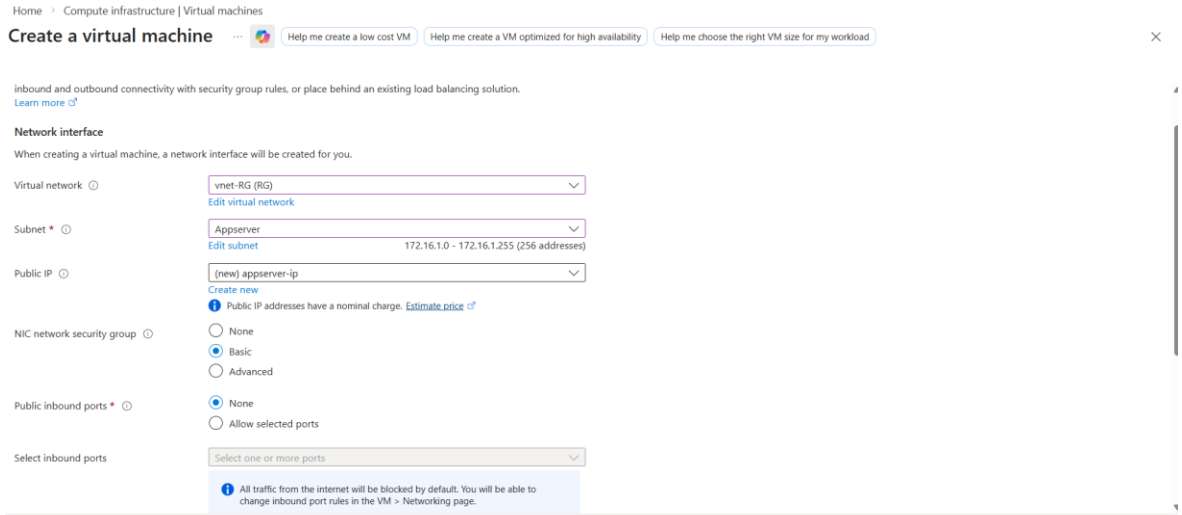


Step 3: Created Web Server VM.
- Disk Type: Standard HDD
- Username: mokesh
- Installed Nginx using custom script:

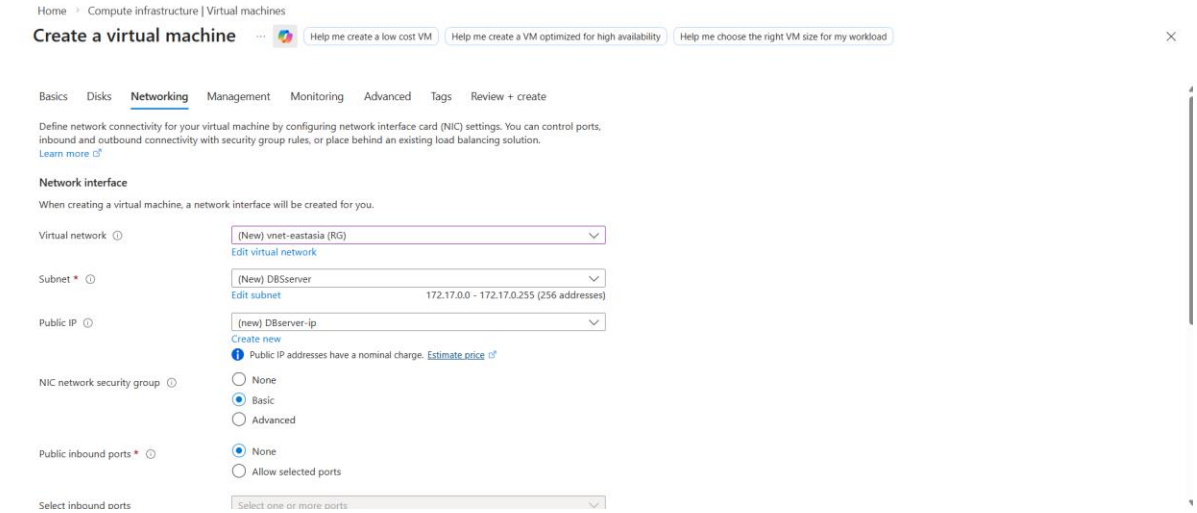
```
#!/bin/bash
sudo su
apt update
apt install nginx -y
```

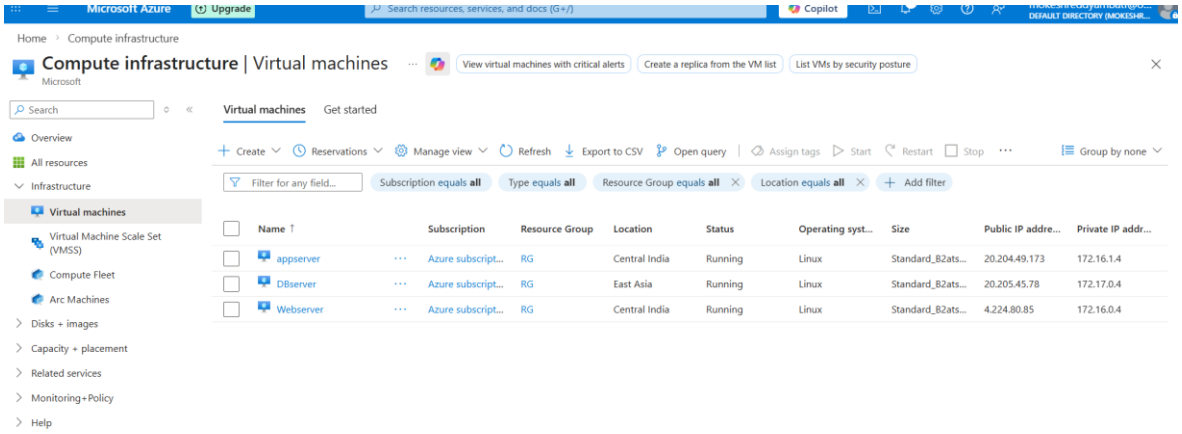


Step 4: Created Application Server VM.
- Selected previously created subnet (App Subnet)
- SSH port disabled (None)



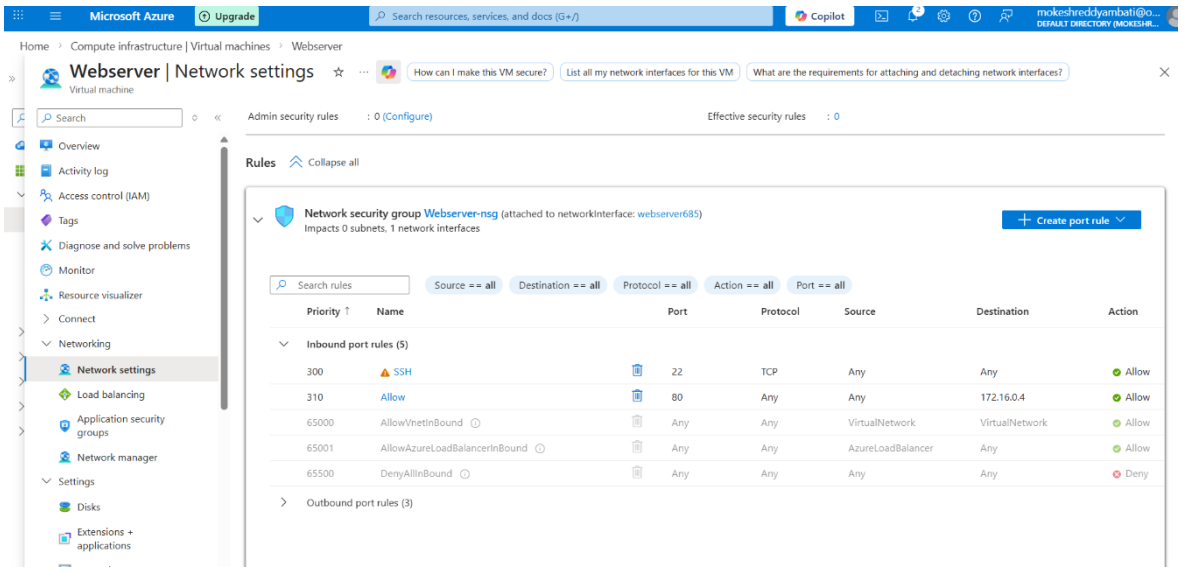
Step 5: Created Database Server VM in East Asia Region.
- Selected DB Subnet
- SSH port disabled (None)

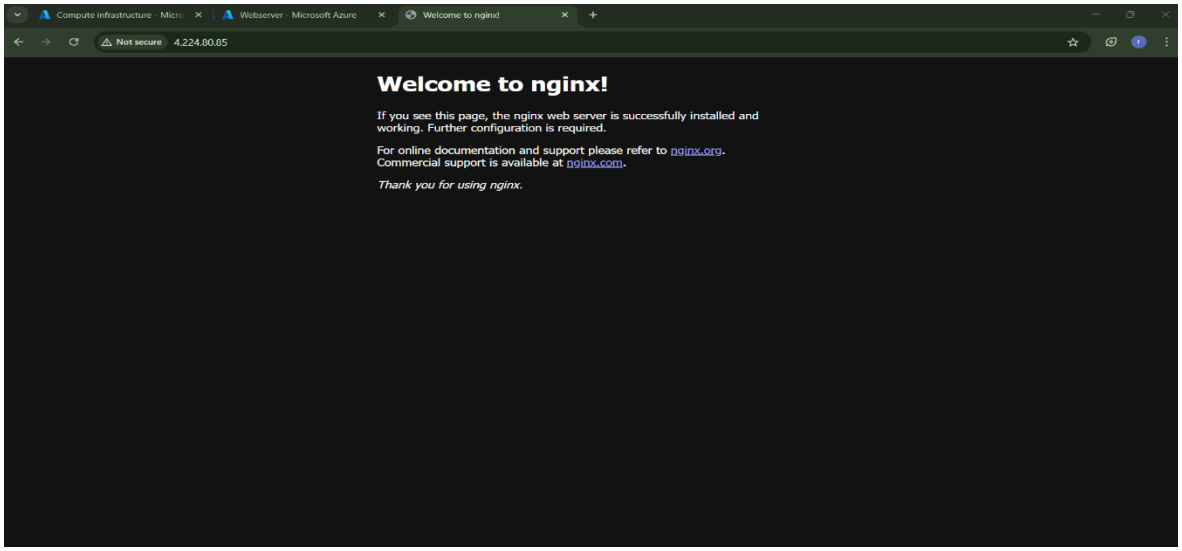




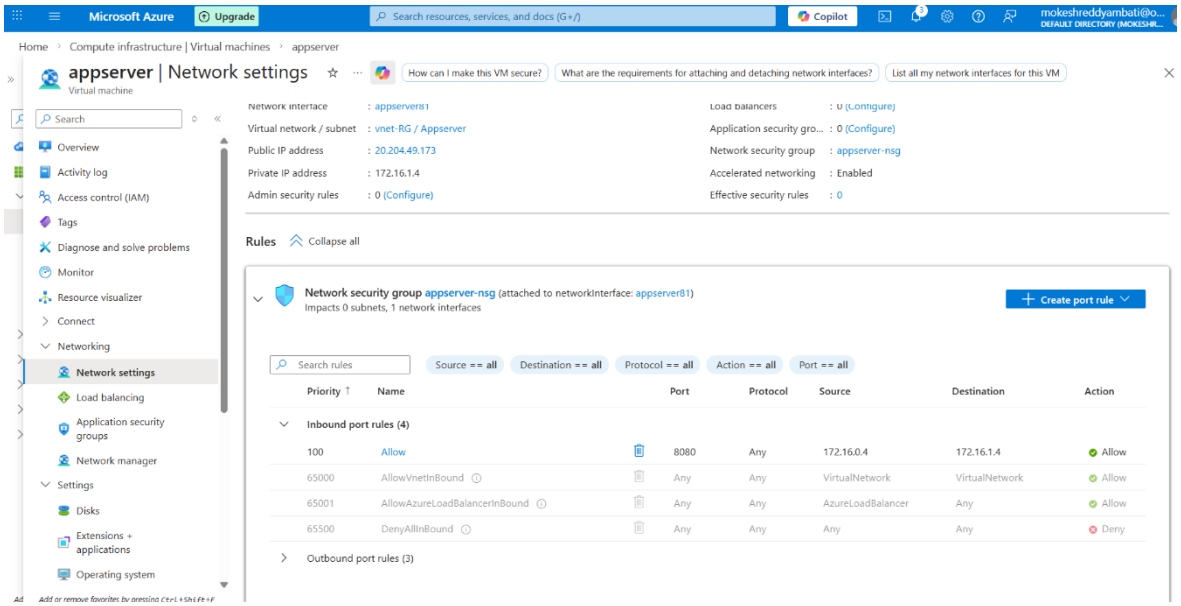
Step 6: Configured Network Security Group (NSG) Rules.

Web Server:
- Allowed HTTP (Port 80) from Any source

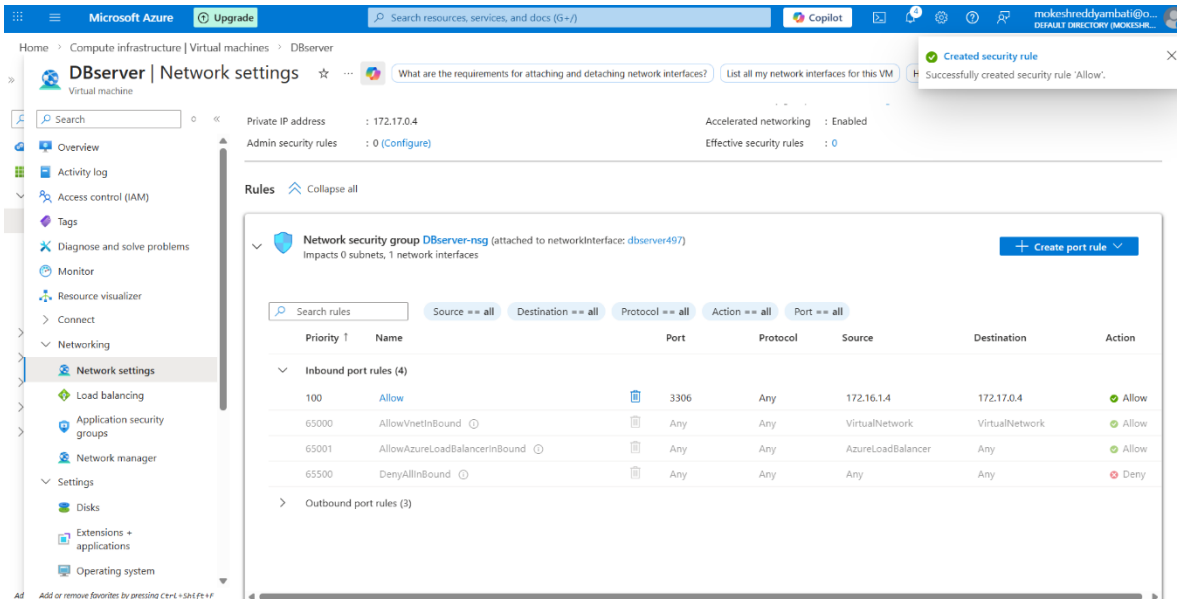




App Server:
- Allowed Port 8080 from Web Server IP



Database Server:
- Allowed Port 3306 from App Server IP



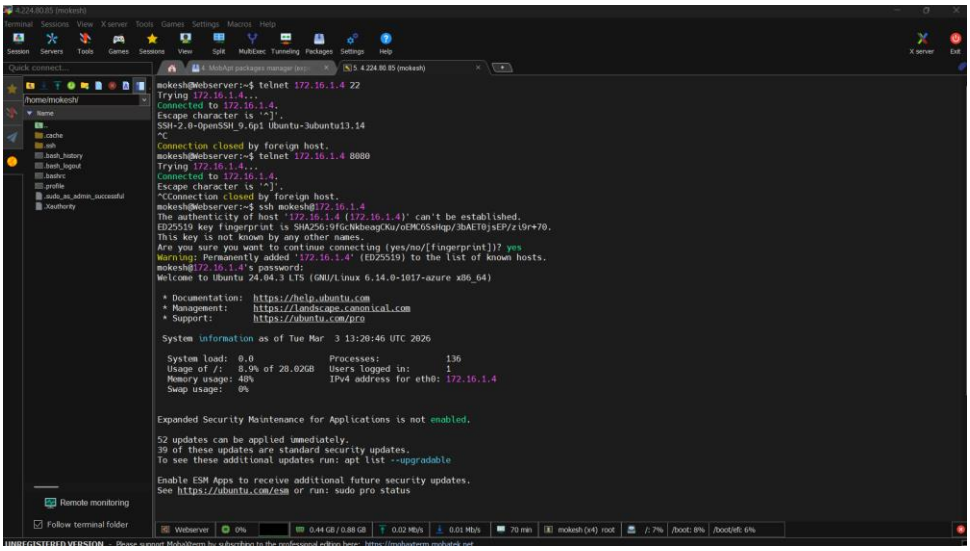
Step 7: VNet Peering configured between regions (Central India and East Asia) to allow communication between VMs in different VNets.

Step 8: Connectivity Testing using MobaXterm and Telnet.

Verified:
- Web to App (Port 22 and 8080)
- App to DB (Port 22 and 3306)

Step 9: Installed Tomcat on App Server.
Commands used:
Ssh username@appserver_private_ip
Check: hostname

Bash:
sudo apt update
sudo apt install openjdk-17-jdk -y
java -version
cd /opt
sudo wget https://archive.apache.org/dist/tomcat/tomcat-11/v11.0.18/bin/apache-tomcat-11.0.18.tar.gz
sudo tar -xvzf apache-tomcat-11.0.18.tar.gz
sudo mv apache-tomcat-11.0.18 tomcat
ls /opt
cd /opt/tomcat/bin
sudo ./startup.sh
sudo ss -tulnp | grep 8080
0.0.0.0:8080



Step 10: Installed MySQL on DB Server.

Commands used:
ssh username@dbserver_private_ip
Check: hostname
Bash:
sudo apt update
sudo apt install mysql-server -y
sudo systemctl status mysql
sudo mysql_secure_installation
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
Find : bind-address = 127.0.0.1
Change To : bind-address = 0.0.0.0
sudo systemctl restart mysql
sudo mysql -u root -p

INSIDE MYSQL:
CREATE DATABASE appdb;

CREATE USER 'appuser'@'%' IDENTIFIED BY 'password123';

GRANT ALL PRIVILEGES ON appdb.* TO 'appuser'@'%';

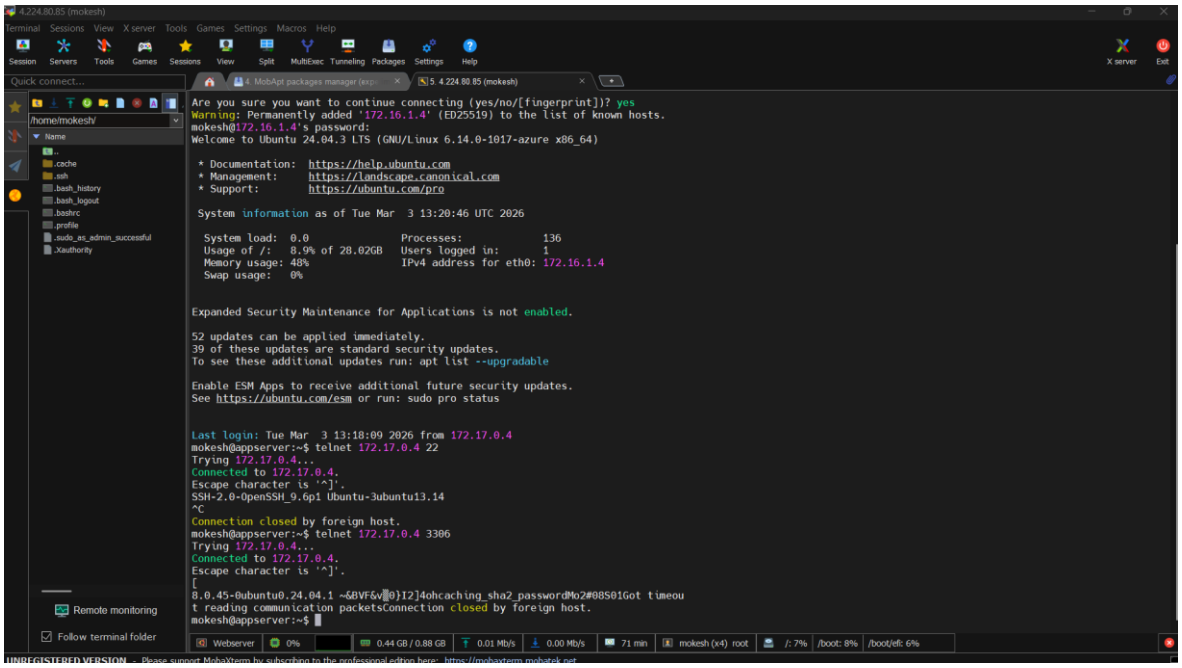
FLUSH PRIVILEGES;

EXIT;

TEST FROM APP SERVER :

mysql -h db_private_ip -u appuser -p

After installation, tested database connectivity using:
telnet 3306



Advantages of 3-Tier Architecture

- Improved Security
- Better Scalability
- Independent Layer Management
- Easy Maintenance
- Secure Database Isolation

Conclusion

Successfully implemented a secure 3-Tier Architecture in Microsoft Azure with Web, Application, and Database layers. Configured NSG rules, VNet Peering, and verified connectivity using Telnet.